Oumaima Chater

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Summary

AI Engineer with a solid foundation in statistics, optimization, and machine learning. Experienced in developing interpretable models, time-series forecasting, NLP, and computer vision. Skilled in building scalable ML pipelines and collaborating with cross-functional teams to align technical solutions with business needs.

Technical Skills

Languages: Python, C++, SQL, MATLAB

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, VAE, LangChain, RAG, LSTM, CNNs, multimodal models, NLP, computer vision

Data & Storage: Pandas, NumPy, SciPy, PostgreSQL, MongoDB

Cloud & MLOps: GCP, AWS Lambda, Vertex AI, FastAPI, GitHub Actions, Power BI

AI Applications: RAG systems, document intelligence, generative models, counterfactual explanations

Experience

AI Engineer Dec 2024 – Present
Libertify *Paris, FR*

- Developed scalable ML pipelines including RAG systems for document/image processing using Vertex AI and AWS Lambda.
- Automated content generation via AI agents, accelerating editorial throughput.
- Built end-to-end pipeline that transformed scripts into production-ready visual assets via generative image models.

AI Intern May 2024 – Oct 2024
Renault *Guyancourt, FR*

- Designed interpretable models for vehicle fault detection, addressing class imbalance with tailored resampling strategies.
- Automated comparison of RFP documents using multimodal LLM and computer vision models.
- Collaborated with risk and business teams to align analytics output with operational KPIs.

Projects

Time Series Anomaly Detection (Genvia) Oct 2023 – Apr 2024
Predictive Maintenance with Domain Shift *Python, VAE, Domain Adaptation*

- Built VAE-based domain adaptation models to predict failures in multi-sensor time series data.
- Achieved 90.8% accuracy with counterfactual-based visualizations to support decision-making.

Multi-Agent Navigation Simulator Mar 2024
Planning & Coordination under Constraints *Python, MAS*

- Simulated collaborative navigation in hazardous environments, reducing task time by 3.5×.

Education

CentraleSupelec Gif-sur-Yvette, FR
Engineering Degree (M.S. equiv.) in Computer Vision & AI *Aug 2021 – Dec 2024*

Preparatory Classes (CPGE) Rabat, MA
MPSI & MP – Math, Physics, Engineering* *Sep 2019 – Jun 2021*

Languages

Arabic: Native **French:** Bilingual Proficiency **English:** Full Professional Proficiency